

REVIEW

GTP Hydrolysis Mechanism of Ras-like GTPases

Guangpu Li^{1*} and Xuejun C. Zhang²

¹Department of Biochemistry
and Molecular Biology
University of Oklahoma Health
Sciences Center, 940 S.L.
Young Blvd, BMSB 853
Oklahoma City, OK 73104
USA

²Crystallography Research
Program, Oklahoma Medical
Research Foundation
Oklahoma City, OK 73104
USA

The Ras-like GTPases regulate diverse cellular functions *via* the chemical cycle of binding and hydrolyzing GTP molecules. They alternate between GTP- and GDP-bound conformations. The GTP-bound conformation is biologically active and promotes a cellular function, such as signal transduction, cytoskeleton organization, protein synthesis/translocation, or a membrane budding/fusion event. GTP hydrolysis turns off the GTPase switch by converting it to the inactive GDP-bound conformation. The fundamental GTP hydrolysis mechanism by these GTPases has generated considerable interest over the last two decades but remained to be firmly established. This review provides an update on the catalytic mechanism with discussions on recent developments from kinetic, structural, and model studies in the context of the various GTP hydrolysis models proposed over the years.

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*Corresponding author

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Introduction

The simple GTP hydrolysis cycle provides a universal switch mechanism in controlling numerous cellular functions through the Ras-like small GTPases^{1–3} as well as other classes of GTPases such as the signaling G_{α} proteins,⁴ signal recognition particle (SRP) GTPases (e.g. SRP54 and SR α)^{5,6} and protein translation elongation factors (e.g. EF-Tu and EF-G).⁷ The Ras-like small GTPases contain more than 100 20–30 kDa proteins that function as molecular switches in regulation of diverse cellular functions, including cell proliferation/differentiation, cytoskeleton organization, and intracellular membrane trafficking.⁸ There are five subfamilies: Ras, Rho, Rab, Arf, and Ran. While they show different intracellular localizations and interact with distinct sets of effectors, accounting for their functional specificity, they all share the same three-dimen-

sional structure, the so-called G domain responsible for GTP/GDP binding and GTP hydrolysis.³ These GTPases have extremely high affinity for GTP/GDP with a K_D usually in the picomolar to nanomolar range, with the exception of SRP GTPases that have lower GDP affinity.⁹ As a result, they alternate between GTP- and GDP-bound conformations in the cell and rarely exist in nucleotide-free state. The GTP-bound conformation is biologically active and turns on a cellular function *via* interaction with and activation of a specific effector(s). In the case of Ras, one of its effectors is the Raf kinase.^{10–13} GTP-bound Ras binds Raf and facilitates the activation of Raf, which in turn activates the MAPK signaling cascade,^{14,15} leading to cell proliferation/differentiation. GTP hydrolysis turns off the GTPase switch by converting it to the inactive GDP-bound conformation that has low affinity for the effector(s). In the cell, the on-off switch of GTPases is further regulated by GTPase-activating proteins (GAP) that accelerate GTP hydrolysis^{16–18} and guanine nucleotide exchange factors (GEF) that accelerate GDP dissociation and reloading of GTP.^{19–21}

The lifetime of the GTP-bound conformation can be viewed as a timer that determines the activation time of a cellular function.^{4,22–24} The biological significance of GTP hydrolysis thus lies in turning off the timer and the cellular function.²⁵ Furthermore, recent mathematical modeling studies suggest that the energy expenditure in GTP hydrolysis can also

Abbreviations used: GAP, GTPase-activating protein; RGS, regulator of G protein signaling; P-loop, phosphate-binding loop; LFER, linear free energy relationship; FTIR, Fourier transform infrared spectroscopy; PIX, positional isotope exchange; ESEEM, electron spin-echo envelope modulation; GppNHP, guanosine 5'-(β,γ)-imidotriphosphate; SRP, signal recognition particle; SR, SRP receptor; ER, endoplasmic reticulum.

E-mail address of the corresponding author:
guangpu-li@ouhsc.edu

improve the timer accuracy by reducing the variance in the lifetime of the active GTP-bound conformation at the level of single GTPase molecules.²³ The importance of GTP hydrolysis in GTPase function is well documented. In the case of Ras, mutations that reduce its GTP hydrolysis rate and thus prolong the GTP-bound conformation and activation time of cell proliferation are frequently found in a variety of cancers.^{26,27} In the case of signaling G_{α} proteins, mutations that block the GTPase activity of $G_{s\alpha}$ and $G_{i\alpha}$ are associated with pituitary tumors.²⁸ In recent years, considerable effort has been directed towards understanding the fundamental mechanism of GTP hydrolysis and this review focuses on these developments.

Dissociative *versus* associative transition state

In general, chemical reactions need to overcome an energy barrier (transition state) in the conversion from substrates to products and the catalytic power of enzymes lies in their ability to stabilize the transition state and thus lower the energy barrier. For phosphoryl transfer reactions, such as GTP hydrolysis, possible transition states include a continuous spectrum of dissociative to associative structures.^{29,30} On one extreme, a fully dissociative transition state is represented by a metaphosphate intermediate that shows extensive

bond cleavage between the γ -phosphoryl group and the GDP leaving group (Figure 1). On the other extreme, a fully associative transition state is represented by a pentacovalent trigonal bipyramidal intermediate that shows no β - γ bridge bond cleavage but significant amount of bond formation between the incoming nucleophilic water molecule and the γ -phosphate of GTP (Figure 1). In these two extremes, there is contrast in charge distribution. In a dissociative transition state, negative charges shift to the β -phosphate with the largest accumulation on the β - γ bridge oxygen, while in an associative transition state, negative charges accumulate on the non-bridging atoms of the γ -phosphate.

Initial GTP hydrolysis models invoked associative transition state.^{31–38} Several models have focused on identifying a general base that would abstract a proton from the nucleophilic water molecule and thus favor the bond formation between the nucleophilic water molecule and the γ -phosphate of GTP in the associative transition state. Originally the conserved Gln residue in the switch II region (the Gln61 of Ras) was considered as the general base,³⁵ but subsequent experiments and theoretical studies have since argued against this model. First, substitution of the Gln61 residue of Ras with an unnatural amino acid NGln, which is an isoelectronic, isosteric nitro analog of Gln and a much poorer base than Gln, has little effect on either intrinsic or GAP-accelerated GTP hydrolysis rate.³⁹ Second, computer simulation studies and analysis of linear free energy relationship (LFER) suggest that Gln61 of Ras cannot serve as the general base for the GTP hydrolysis reaction.^{34,38} Current general base models suggest that the substrate GTP itself, specifically the γ -phosphate of GTP, is the general base in its own hydrolysis.^{37,40,41} These models are based on theoretical calculations of LFER between the GTPase reaction rate and the pK_a of the γ -phosphate of GTP⁴⁰ as well as experimental observations that the GTPase reaction rate decreases when the pH drops below 3, the calculated pK_a of the bound GTP γ -phosphate.³⁷ However, a recent vibrational spectroscopy study has raised the issue if this mechanism actually works in GTPase-catalyzed hydrolysis reactions, with the observation that the pK_a of the bound GTP γ -phosphate is below 3.3 and there is no evidence of simple protonation of the bound GTP.⁴²

Furthermore, other models are proposed to emphasize the contribution of a GTPase active site to the stabilization of the transition state.^{31,36} In Ras, the GTP-binding pocket contains positive charges and hydrogen bond donors, including Mg^{2+} , Lys16, and the backbone amide groups of the P-loop. Ras GAP contributes another positively charged Arg (the so called Arg finger) to the active site⁴³ to accelerate the hydrolysis reaction. These functional groups interact with both γ - and β -phosphate groups of GTP in GDP + $AlF_{3/4}$ -bound, transition-state mimic structures. These

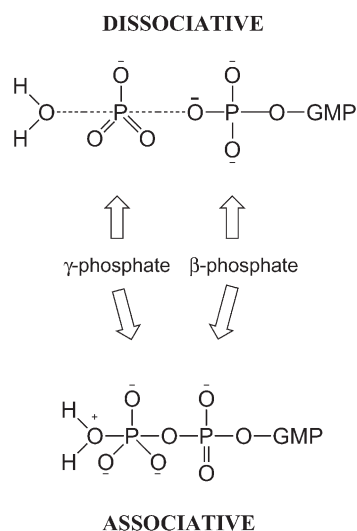


Figure 1. Dissociative *versus* associative transition state. In GTP hydrolysis, a dissociative transition state exhibits extensive bond cleavage between the γ - and β -phosphate groups, with the shift of negative charges to the β -phosphate at the β - γ bridge oxygen as well as the two non-bridging oxygen atoms. In contrast, an associative transition state shows no bond cleavage between the γ - and β -phosphate groups, with the shift of negative charges to the γ -phosphoryl oxygen atoms. Partial charges are indicated for the bridging and non-bridging oxygen atoms in both transition states.

models emphasize the stabilization of the γ -phosphate in an associative transition state, although the interactions with the β -phosphate can favor a dissociative transition state (see below). For the signaling G_{α} subunits, the catalytic Arg residue is provided in *cis* (e.g. Arg178 of $G_{i\alpha 1}$). In these models, the conserved switch II Gln residue (Gln61 of Ras or Gln204 of $G_{i\alpha 1}$) is suggested to stabilize the transition state by orientating the nucleophilic water molecule and/or stabilizing the charge development at the γ -phosphate of GTP in an associative transition state.^{31,44} Orientation of the nucleophilic water molecule, however, can also be catalytic in a dissociative reaction (see below).

More recent GTP hydrolysis models suggest that the transition state is dissociative in nature. GTP hydrolysis in solution forms a dissociative transition state, as shown by model studies.⁴⁵ Maegley *et al.* have proposed that GTP hydrolysis by a GTPase may follow the same dissociative reaction pathway,²⁹ with the observation that the nature of the transition state for phosphoryl transfer reactions is often difficult to change.^{46,47} Indeed, recent kinetic and structural studies using GTPase-defective mutants, X-ray crystallography, Fourier transform infrared spectroscopy (FTIR) and Raman difference spectroscopy have provided evidence in support of a mostly dissociative mechanism for GTP hydrolysis by Ras-like GTPases.^{48–53} Several research groups have employed FTIR and Raman spectroscopy studies to determine the changes in the vibrational mode of γ -PO₃ and β -PO₂ groups of GTP upon Ras binding, which has implications on the interactions, bond order, and charge shift at these functional groups.^{48–52,54} It is important to note that their spectra assignments are in general agreement. A key feature of these vibration spectroscopy studies is that the vibration of P_{γ} -O_{3 non-bridge} and P_{β} -O_{2 non-bridge} bonds are independent and are not coupled to other P-O backbone stretches in GTP, which allows the determination of interactions and implied charge shift at the γ -PO₃ and β -PO₂ groups. These studies have shown a downshift of the β -PO₂ but not γ -PO₃ stretch of GTP upon its binding to Ras, indicating strong interactions between Ras and the β -PO₂ of GTP. These interactions suggest a decrease in the P_{β} -O_{non-bridge} bond order and imply a shift of negative charges to the β -phosphoryl oxygen atoms. Importantly, the GAP that stimulates Ras GTPase activity can further enhance this negative charge shift from γ - to β -phosphate.⁴⁸ These results support but not prove a dissociative mechanism. Further evidence for the dissociative mechanism comes from a recent isotope-edited FTIR study, which combines Raman and IR difference spectroscopy to assign both symmetric and anti-symmetric vibrational modes of GDP and GTP upon binding to Ras.⁵⁴ This study suggests that GTP begins to acquire transition state features upon binding to Ras, i.e. the γ -phosphate approaches a more planar

geometry and the bridging P-O bond is weakened. This bridging bond order decrease correlates with the Ras-mediated increase in GTP hydrolysis rate, consistent with the classical dissociative phosphoryl transfer reactions in solution.^{45,55}

The two non-bridging oxygen atoms of β -PO₂ show extensive hydrogen bonding and electrostatic interactions with the P-loop residues (the backbone amide groups of Val14, Gly15, and Lys16) and Mg^{2+} , respectively, at the active site of Ras crystal structure.^{35,36} NMR data also suggest strong hydrogen bonds between the amide groups of Gly13 and Lys16 in the P-loop and the β -phosphate of bound GDP and GTP analogs.^{56–59} These interactions can stabilize the negative charge build-up at β -PO₂ in a dissociative transition state and thus may play a catalytic role in Ras-catalyzed GTP hydrolysis.

A fully dissociative transition state is a metaphosphate structure. However, a discrete free metaphosphate structure is not absolutely required for a phosphoryl transfer reaction to proceed *via* a dissociative mechanism in solution.^{46,55} In an effort to identify such a structure in Ras-catalyzed GTP hydrolysis reaction, time-resolved FTIR has been used to monitor the positional isotope exchange (PIX) of [β -¹⁸O₃]GTP during the GTP hydrolysis reaction by Ras.⁵¹ If PIX occurs, then there would be a reversible cleavage of the P_{γ} -O _{β - γ bridge} bond in GTP, and thus a metaphosphate intermediate. However, no PIX was detected in these experiments, suggesting that there is no discrete metaphosphate intermediate during the Ras-catalyzed GTP hydrolysis reaction. Additional oxygen isotope exchange measurements have since been performed with Ras-[γ -¹⁸O₃]GTP in the presence of a Ras GAP (NF1) and again there is very little isotope exchange.⁶⁰ These data are consistent with a concerted mechanism⁵² involving a transition state that is mostly dissociative in nature but does not form a free metaphosphate.

A dissociative transition state requires a shift of negative charges to the β -phosphate of GTP, including the β - γ bridge oxygen and the two non-bridging oxygen atoms (Figure 1). The FTIR studies have confirmed strong Ras interactions at the two non-bridging β -phosphoryl oxygen atoms, which may be interpreted as negative charge shift to these atoms. However, this technique has its limitations and is unable to monitor accurately the interaction and bond changes at the β - γ bridge oxygen, because the vibration of the P_{γ} -O _{β - γ bridge} bond is coupled to other P-O backbone stretches. To resolve this issue, additional techniques are necessary. In this regard, the kinetic isotope effect that measures the charge difference between the transition state and the ground state is perhaps the most promising. Recently, ¹⁸O kinetic isotope effects have been determined in Ras-catalyzed GTP hydrolysis and the results show that there is significant kinetic isotope effect on the β - γ bridge oxygen (X. Du, personal communication), providing further evidence for a largely dissociative transition state.

It has been proposed that the active site group(s) of GTPases involved in stabilizing the charge build-up at the β - γ bridge oxygen may play a catalytic role in GTP hydrolysis.²⁹ In the case of Ras, that catalytic group is the backbone amide group of Gly13 in the P-loop. Recent evidence supporting a catalytic role for this backbone amide group in the GTP hydrolysis reaction has come from kinetic and structural studies of Rab5,^{53,61} a Ras-like small GTPase involved in endocytic trafficking. In Rab5, the equivalent P-loop residue is Ala30 (Figure 2). Nineteen substitution mutations have been introduced at this residue and only the Pro substitution, which changes the backbone amide to an imide group that can no longer donate hydrogen bond, drastically reduces GTP hydrolysis rate by an order of magnitude.⁶¹ In fact, this Rab5 : A30P mutant shows such a low GTPase activity that it forms a GTP-bound complex suitable for structural studies *via* X-ray crystallography.⁵³

Rab5 : A30P in complexes with GTP, GDP, as well as GDP + AlF₃ have been crystallized and

high resolution crystal structures are obtained at 1.60 Å, 1.60 Å, and 1.55 Å, respectively⁵³ (Figure 2). The A30P mutation does not cause global conformational changes, instead the structural changes are mainly restricted at the mutation site affecting the peptide plane of residues 29 and 30 (compare Figure 2(A) and (B)), while the switch I loop shifts outward by about 0.5 Å. Other P-loop residues such as Val31, Gly32, and Lys33 are not affected, neither is the overall conformation of the P-loop⁵³ (Figure 2(A) and (B)). Thus, the defective GTP hydrolysis by Rab5 : A30P is most likely due to the loss of a catalytic component, i.e. the backbone amide group of Ala30, supporting the idea that the backbone amide group of Ala30 (or Gly13 of Ras) in the P-loop plays a catalytic role in the GTP hydrolysis reaction by forming a hydrogen bond with and stabilizing the β - γ bridge oxygen where the largest charge build-up occurs in a dissociative transition state. The (GDP + AlF₃)-bound structure of Rab5 : A30P mimics the transition state where the side-chain of the catalytic Gln79 (or Gln61 of Ras) in the switch II region dips into the

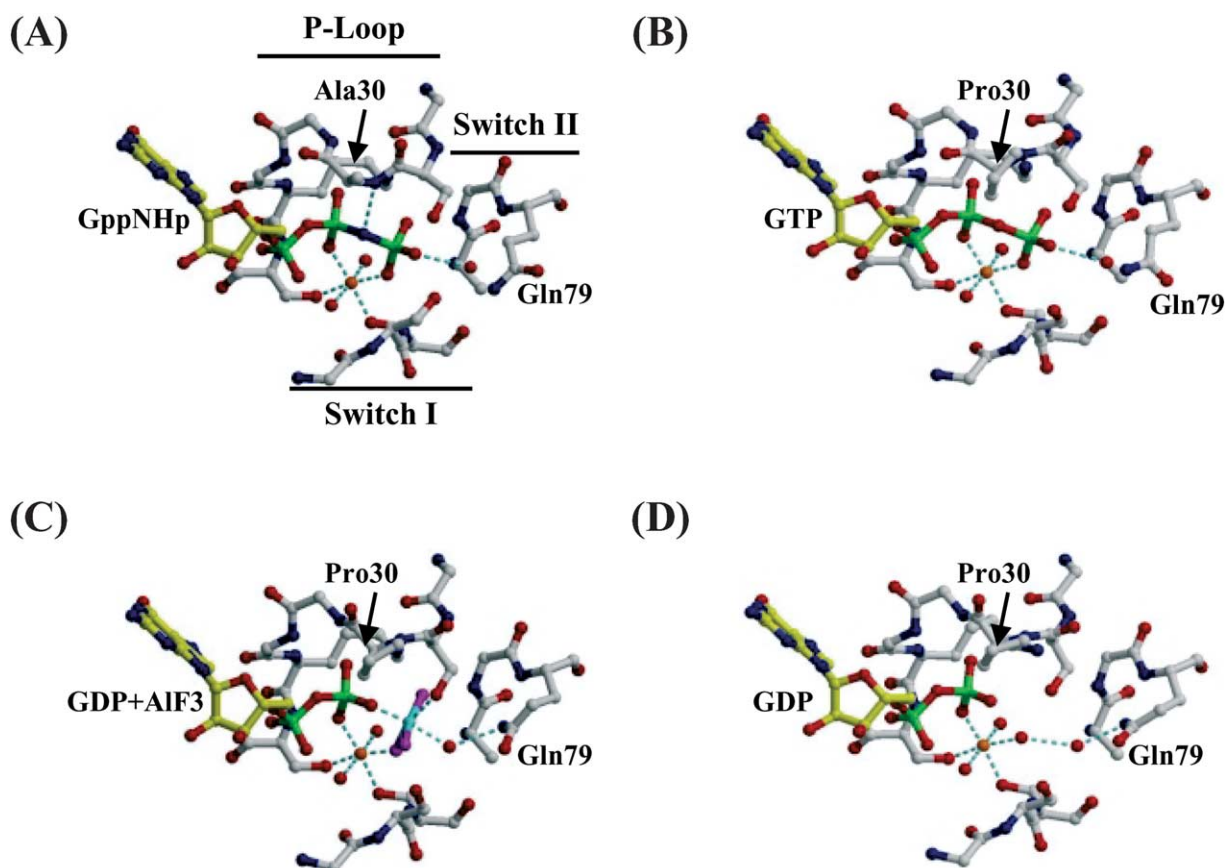


Figure 2. X-ray crystal structures of Rab5 and the GTPase-defective A30P mutant. (A) Wild-type complexed with GppNHp, a non-hydrolyzable GTP analog (PDB code 1N6H). (B) A30P mutant complexed with GTP (PDB code 1N6L). (C) A30P mutant complexed with GDP + AlF₃ (PDB code 1N6K). (D) A30P mutant complexed with GDP (PDB code 1N6I). The P-loop and the switch regions I and II indicated in (A) occupy the same space in other panels. The bound nucleotides, the mutation site (Ala30 or Pro30), the catalytic residue Gln79, and the nucleophilic water molecule are indicated. In the ball-and-stick model, the amino acid backbone and side-chains are shown with carbon in gray, oxygen in red, and nitrogen in blue. The bound nucleotides and derivatives are depicted with phosphorus in green, oxygen in red, nitrogen in blue, aluminum in light blue, fluoride in pink, and the rest in yellow. Other active site groups include magnesium ion and water molecules, which are shown as gold and red balls, respectively. Dotted lines indicate hydrogen bonds.

nucleotide-binding pocket and forms hydrogen bond with the nucleophilic water molecule to position it correctly for attacking the γ -phosphate (represented here by the AlF_3 group)⁵³ (Figure 2(C)). The catalytic conformation of Gln79 and the nucleophilic water position are maintained in the GDP-bound structure, suggesting that the side-chain of Gln79 is sufficient to align the water molecule without the need of the γ -phosphate of GTP⁵³ (Figure 2(D)). Thus, these crystallographic data on Rab5 corroborate the kinetic isotope effect and FTIR data on Ras and together with the electron spin-echo envelope modulation (ESEEM) and molecular dynamics simulation data (see the next section) support the dissociative transition state model. While the GTP-bound Rab5 : A30P structure confirms the low GTPase activity of this mutant, the implied GTP hydrolysis mechanism and the GDP- and GDP + AlF_3 -bound mutant structures need to be interpreted with caution. For example, the A30P mutation also causes a 0.5 Å outward shift of the switch I region, which could contribute to the decreased GTPase activity to some extent. Although the Rab5 : A30P structures were solved with bound nucleotides, biochemical studies have shown that the Rab5 : A30P mutant decreases binding to GTP and GDP in solution,^{61,62} and possibly to GDP + AlF_3 , like other GTP hydrolysis-defective mutants.

Catalytic groups and GTPase-activating proteins

The catalytic groups in GTPase-catalyzed GTP hydrolysis reactions are those that can stabilize the negative charge build-up at the β - or γ -phosphate of GTP, depending on a dissociative or an associative mechanism, as well as correctly position the nucleophilic water molecule at the transition state. Orientation of the nucleophilic water molecule can catalyze both dissociative^{63–65} and associative^{31,43,44} hydrolysis reactions. In this regard, the conserved switch II Gln residue in GTPases is suggested to contribute to catalysis by orientating the nucleophilic water molecule for in-line attack of the γ -phosphate of GTP.^{31,43,44,53,65–67} In both small Ras-like GTPases and G_α subunits, the side-chain of the catalytic Gln residue only exhibits catalytic geometry at the transition state but not at the ground state.^{31,43,53,68,69} Figure 2 illustrates the situation with Rab5 and Rab5 : A30P mutant structures as examples. The side-chain of the Gln79 residue in Rab5 (or Rab5 : A30P) is not in a geometry to position correctly the nucleophilic water molecule at ground GTP-bound state⁵³ (Figure 2(A) and (B)). There is a “weak” 3.3 Å hydrogen bond between the Gln and the water molecule in the GTP-bound form, but the water molecule is not aligned correctly for catalysis. At GDP- and GDP + AlF_3 (transition state mimic)-bound state, a conformational change occurs in the backbone of the switch II region so that the

Gln side-chain tip plunges into the GTP-binding pocket and repositions the nucleophilic water molecule (Figure 2(C) and (D)). As a result, the nucleophilic water molecule is pushed towards the γ -phosphate atom by 1.5 Å and correctly aligned along the $\text{P}_\gamma\text{--O}_{\beta\gamma}$ bridge axis for the nucleophilic attack (Figure 2).

In addition to the aforementioned Gln residue in positioning the nucleophilic water molecule, another catalytic group is suggested to be the positively charged Arg residue (called Arg finger) that stabilizes the negative charge development at the phosphate groups of GTP during the hydrolysis reaction.^{31,43,68,70,71} For example in $G_{i\alpha 1}$, it is the Arg178.³¹ Such a catalytic Arg residue cannot be found in the structures of Ras-like small GTPases, but it can be provided in *trans* by their GAPs,^{43,68,69} e.g. Arg789 of the Ras GAP.^{17,72} In the transition state structures of both Ras-like small GTPases and G_α subunits, this catalytic Arg contacts the γ -phosphate analog as well as the β – γ bridge oxygen.^{31,41,43,68,69} The associative model emphasizes the interaction with and stabilization of the γ -phosphate,^{31,33,43} while the dissociative model emphasizes the interaction with the β – γ bridge oxygen and suggests that the interaction with the γ -phosphate is not catalytic but may facilitate binding and orientation of the γ -phosphate during the reaction.²⁹ In addition to charge neutralization, the backbone carbonyl group of the Arg finger also helps the switch II Gln position the nucleophilic water molecule and thus stabilizes the transition state.^{69,73} Furthermore, by comparison between the crystal structures of ground state and transition state Rho GTPases complexed with the Rho GAP, it is found that the Arg finger only assumes a catalytic position at the transition state, suggesting that for the Ras-like small GTPases, the GTPase–GAP complex needs to undergo conformational changes to bring the Arg finger to the active site for catalysis.⁶⁹

The intrinsic GTPase activity of small Ras-like GTPases is usually lower than the G_α subunits because of lacking the catalytic Arg finger, but is still much higher than the GTP hydrolysis rate in solution. According to the dissociative model, one important catalytic group is the backbone amide group of Rab5 Ala30 in the P-loop (or the equivalent position in other GTPases, e.g. Gly13 of Ras). In addition to the structural data on the Rab5 : A30P mutant discussed in the previous section, earlier solution NMR studies suggest that the backbone amide group of Ras Gly13 forms strong hydrogen bond with the β – γ bridge atom of GppNHp (a non-hydrolyzable GTP analog) as well as the β -phosphate of GDP.^{56–58} Furthermore, recent ESEEM and molecular-dynamics simulation studies reveal movement of the amide group of the Gly13 residue of Ras towards the metal ion in the active site upon binding to the GAP, supporting its catalytic role in the GAP-accelerated GTPase reaction *via* stabilization of the β – γ bridge oxygen of GTP.⁷⁴ In addition to the β – γ bridge oxygen, the

two non-bridging β -phosphoryl oxygen atoms of GTP also accumulate negative charges in a dissociative transition state,^{48–51} despite at a smaller scale.²⁹ Thus, groups that interact with and stabilize the two non-bridging oxygen atoms can also be catalytic. These catalytic groups include Mg^{2+} and the other backbone amide groups of the P-loop at Val31, Gly32, and Lys33 of Rab5 (or Val14, Gly15, and Lys16 of Ras or equivalent residues of other GTPases), highlighting the importance of P-loop backbone conformation in intrinsic GTPase activity. It should be noted that while Mg^{2+} is important for GTP hydrolysis, it is already bound to GTP in solution and may not be a major factor for further rate enhancement in GTPase-catalyzed reactions. From the perspective of the associative model, however, the intrinsic GTPase activity involves using the substrate GTP as a general base in the catalysis.^{37,40,41}

The low intrinsic GTP hydrolysis rate of Ras-like GTPases is subject to up-regulation by GTPase-activating proteins (GAPs) in the cell,^{16–18} which provides a physiological mechanism in regulating the dynamics of the GTPase switches and the diverse cellular functions controlled by the various GTPases. Rab5 is among the fastest small GTPases with an intrinsic k_{cat} of $2 \times 10^{-1} \text{ min}^{-1}$,⁷⁵ while Ras has a k_{cat} of $2.8 \times 10^{-2} \text{ min}^{-1}$.⁷⁶ The low intrinsic rate may reflect the relatively weak catalytic groups such as the P-loop backbone amide groups, which are not as strong hydrogen bond donors as positively charged side-chains. In addition, the catalytically important Gln side-chain is not in a proper conformation at ground state but needs adjustment to position correctly the nucleophilic water molecule at transition state. This may also contribute to the low intrinsic GTP hydrolysis rate. Ras and Rho GAPs greatly accelerate the GTP hydrolysis rate up to five orders of magnitude⁷⁷ by providing a strong hydrogen bond donor (the Arg finger) to the nucleotide-binding pocket as well as stabilizing the catalytic conformation of the Gln side-chain in the switch II region.^{43,68,69} The Arg finger from the GAPs is in a position to form hydrogen bonds with the β - γ bridge oxygen of GTP as well as a γ -phosphoryl oxygen (or the equivalent fluoride atom in the $GDP + AlF_{3/4}$ -bound structures)^{43,68,69} (Figure 3(A) and (B)). In a dissociative reaction, the interaction with the β - γ bridge oxygen would be catalytic, while in an associative reaction, the interaction with the γ -phosphate would be catalytic. Recent FTIR data that show GAP-mediated enhancement of negative charge shift to the β -phosphate appear more consistent with the dissociative model.⁴⁹ In addition to the Ras and Rho GAPs, Rab GAPs are suggested to work in a similar fashion, based on the GAP domain structure of Gyp1,⁷⁸ but confirmation of the catalytic Arg finger would require the elucidation of Rab-GAP complex structures, which are not yet available.

In contrast to the Ras and Rho GAPs, there are no catalytic Arg residues in Ran and ARF

GAPs^{65,66} (Figure 3(C)). Instead, they mainly rely on the stabilization of the switch II region and reorientation of the catalytic Gln residue as the mechanism to enhance Ran and ARF GTPase activities.^{65,66} This mechanism is similar to the one used by RGS proteins in stimulating the GTPase activity of G_{α} subunits⁷⁹ (see below). Thus, a conserved mechanism used by GAPs to enhance GTPase activity is to interact with the switch II region and reorient the catalytic Gln residue, which in turn aligns the nucleophilic water molecule in geometry to attack the γ -phosphate of GTP. One exception to this rule is the Rap GAPs, as suggested in a recent report describing the crystal structure of the Rap1GAP.⁸⁰ Rap1GAP does not contain an Arg finger and does not stabilize a catalytic Gln residue either, since Rap1 is a GTPase that does not have the catalytic Gln residue. Instead, Rap1GAP appears to provide a catalytic Asn residue to align the nucleophilic water molecule for GTP hydrolysis, thus mimicking the catalytic function of the Gln residue in other GTPases.⁸⁰

The signaling G_{α} subunits exhibit about 100-fold higher intrinsic GTPase activity, because they contain a built-in catalytic Arg residue in their protein sequence, i.e. the catalytic Arg finger is provided in *cis*.⁸¹ Yet the intrinsic GTPase activity can be further enhanced by two orders of magnitude by the so-called RGS proteins (regulators of G protein signaling),^{82–84} which are thus GAPs for the G_{α} subunits. Based on the crystal structure of $G_{i\alpha 1}$ -AlF₄-RGS4 complex, it is suggested that the major catalytic contribution of the RGS protein is the stabilization of the $G_{i\alpha 1}$ switch regions at transition state.⁷⁹ Consequently, the catalytic Gln residue (Gln204 of $G_{i\alpha 1}$) in the switch II region can better position the nucleophilic water molecule for attacking the γ -phosphate of GTP in the hydrolysis reaction.⁷⁹ While the RGS proteins do not donate a catalytic Arg residue, they do share one aspect of the catalytic mechanism with the small GTPase GAPs, i.e. stabilization of the switch II Gln residue and correct positioning of the nucleophilic water molecule.

A new twist in the GTP hydrolysis mechanism comes from the recent elucidation of the crystal structure of the SRP and SR (SRP receptor) complex involved in targeting nascent proteins to the ER membrane of eukaryotic cells or to the cell membrane of bacteria.^{5,6} SRP54 (or Ffh in bacteria) binds to the signal peptide of nascent proteins emerging from the ribosome and then binds to SR (or FtsY in bacteria) on the membrane, which brings the nascent protein to the proximity of the membrane translocon. Once the nascent protein is transferred to the translocon, the SRP and SR complex dissociates and each component is recycled for additional rounds of membrane targeting of nascent proteins. Both SRP and SR are GTPases with a Ras-like GTPase domain and their association-dissociation is regulated by the GTPase cycle. They bind to each other in the

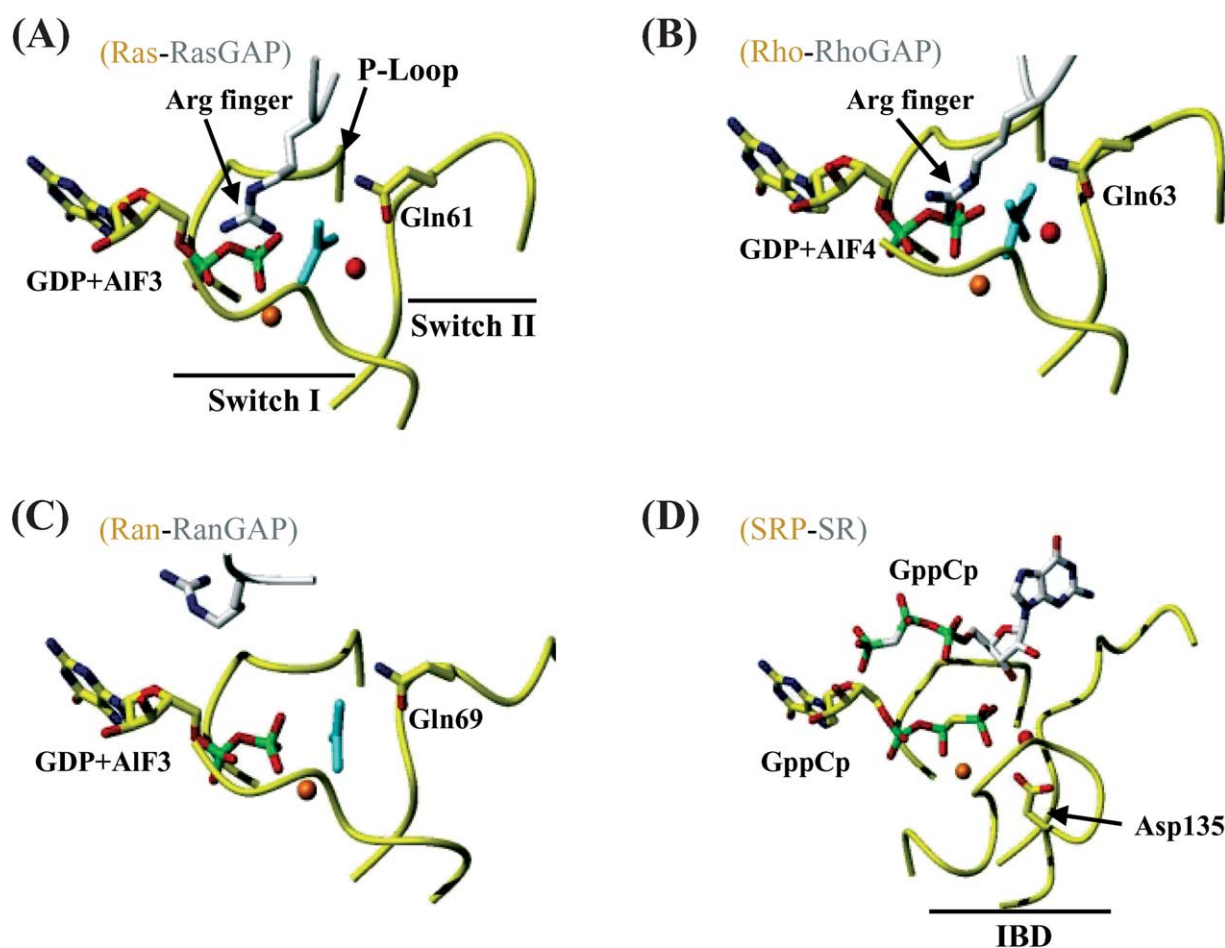


Figure 3. Comparison of GAP-accelerated GTP hydrolysis mechanisms of Ras, Rho, Ran, and SRP. Shown are active site structures of these GTPases in complexes with their GAPs. (A) Ras with bound GDP + AlF_3 in the Ras–RasGAP complex (PDB code IWQI). (B) Rho with bound GDP + AlF_4 in the Rho–RhoGAP complex (PDB code ITX4). (C) Ran–RanGAP complex with bound GDP + AlF_3 (PDB code IK5G). (D) Ffh (ortholog of eukaryotic SRP) with bound GppCp in the Ffh–FtsY (ortholog of eukaryotic SR) complex (PDB code IRJ9). The P-loop and the switch regions I and II indicated in (A) occupy the same space in (B) and (C). In (D) the IBD loop of SRP is indicated. The bound nucleotides, the Arg finger, and the catalytic Gln (A)–(C) or Asp (D), are also indicated. Note that only the Ras and Rho GAPs contain the catalytic Arg finger while the Ran GAP and SR protein do not have one. The backbone and side-chain carbon atoms are shown in yellow for the GTPases and in gray for the GAPs. Other amino acid side-chain atoms are shown with oxygen in red and nitrogen in blue. The bound nucleotides and derivatives are depicted with phosphorus in green, oxygen in red, nitrogen in blue, $\text{AlF}_{3/4}$ in light blue, and carbon in yellow. In (D) there are two bound GppCp molecules: one from SRP and the other one from the complexed SR. To distinguish the two GppCp molecules, the one from SR is shown with its carbon atoms in gray. Other active site groups include the magnesium ion and the nucleophilic water molecule, which are shown as gold and red balls, respectively.

GTP-bound state. Remarkably, the SRP–SR association stimulates the GTPase activity of both proteins, leading to rapid GTP hydrolysis and subsequent dissociation of the complex.^{85,86} In other words, SRP and SR are actually GAPs for each other's GTPase activity.

The crystal structure of the SRP and SR complex reported earlier this year, by two independent groups, has provided insights into the mechanism of this unique reciprocal GTPase activation.^{5,6} The SRP and SR GTPases form a quasi-2-fold symmetrical heterodimer, surrounding the two bound GTP molecules, which are aligned in the symmetrical composite active site so that the 3' OH group of the ribose ring of one GTP molecule contributes a

hydrogen bond to the γ -phosphate of the other GTP molecule and *vice versa* (Figure 3(D)). Thus, the twinned substrate contributes a hydrogen bond *in trans* to the γ -phosphate of the other bound GTP molecule to facilitate complex formation and GTP hydrolysis. In addition, the so-called insertion box domain (IBD) of either protein, which is a unique feature of the SRP and SR GTPases, undergoes conformational changes to position three catalytic residues (an Asp, an Arg, and a Gln) in geometry for GTP hydrolysis (Figure 3(D)). In this case, the Asp positions the nucleophilic water molecule for in-line attack of the γ -phosphate of GTP while the Arg and Gln side-chains interact with the β - and γ -phosphate

groups of GTP to stabilize the charge development during GTP hydrolysis. These catalytic Asp, Arg, and Gln residues interact with the bound GTP *in cis*. Thus, the SRP–SR complex forms a composite active site where two bound GTP molecules catalyze each other's hydrolysis *in trans* with additional rate enhancement from the catalytic residues *in cis* (Figure 3(D)). GTP hydrolysis abolishes the twinned interactions and leads to the dissociation of the SRP–SR complex, providing a structural basis for the coupling of the GTP hydrolysis cycle to the SRP–SR association–dissociation cycle in the membrane targeting of nascent proteins.^{5,6}

Biological functions of GTPase-activating proteins

A common characteristic of GTPases is that their GTP hydrolysis rate is accelerated in the cell by interacting with specific GAPs.¹⁶ It is generally accepted that an important function of GAPs is to serve as negative regulators of GTPases by shortening the lifetime of biologically active GTP-bound conformation. However, GAPs are often large proteins with multiple modular domains. The p120 RasGAP, for example, contains SH2, SH3, PH, and calcium/lipid-binding domains for interactions with receptor tyrosine kinases,^{87,88} p190 RhoGAP,⁸⁹ and membrane lipids,⁹⁰ in addition to its C-terminal GAP domain. The RasGAP, as the first discovered GAP, was also suggested early on to be a downstream Ras effector transmitting the Ras signal, since it interacts with GTP-bound Ras at the so-called effector binding site where other Ras effectors bind to.^{91,92} At structural level, the Arg finger loop of the RasGAP, which is important for catalysis, also appears involved in specific interactions with the Ras proteins.⁹³ However, there is still no evidence to show clearly that the RasGAP can function as a direct downstream Ras effector in a signaling pathway. In this regard, an RGS-containing protein (p115RhoGEF) appears to function both as a GAP and as an effector for the signaling $G_{\alpha 13}$ subunit.^{94,95} Furthermore, phospholipase C and cGMP phosphodiesterase, well known effectors of $G_{\alpha q}$ and transducin, respectively, have been shown to contain GAP activity towards their respective G_{α} subunits.^{96,97}

The concept that GAPs are both negative regulators and effectors of GTPases is further supported by recent experiments on the Arf–ArfGAP and the SRP–SR systems. Arf proteins play a critical role in recruiting coat proteins to the membrane to produce transport vesicles in the intracellular membrane trafficking system. Sar1 is the Arf that is responsible for the formation of COPII-coated vesicles from the ER membrane. GTP-bound Sar1 sequentially recruits the COPII components: Sec23/24 complex and Sec13/31 complex onto membranes/liposomes.⁹⁸ Interestingly, the Sec23 protein is a Sar1 GAP whose

activity is further enhanced upon interaction with the Sec13/31 complex,²² thus the COPII coat has both effector and GAP functions for Sar1. The effector function promotes COPII coat assembly and vesicle formation while the GAP function promotes GTP hydrolysis of Sar1 and COPII coat disassembly. Recent light-scattering studies elegantly demonstrate that the build-in Sar1 GAP activity in the COPII coat makes the coat assembly and disassembly process highly dynamic and self-regulatory.²² Another well-characterized Arf is Arf1, which recruits the COPI coat to the *cis*-Golgi membrane to form COPI-coated vesicles. In this case, the ArfGAP1 is not part of the COPI coat, but it binds to the switch II region of Arf1 to catalyze the GTP hydrolysis by Arf1 and the COPI coat has a different binding site on Arf1.⁶⁶ Interestingly, the COPI coat can work together with ArfGAP1 to enhance further the GTP hydrolysis rate of Arf1.⁶⁶ Furthermore, the membrane curvature can increase the ArfGAP1 activity such that GTP-bound Arf1 recruits COPI on the relatively flat Golgi membrane to form coated buds. With increased curvature in the bud, the ArfGAP1 activity increases and reaches the highest level in the newly formed vesicles, leading to rapid GTP hydrolysis of Arf1 and subsequent coat disassembly.⁹⁹ Similar to the Arf–ArfGAP situation, the newly reported SRP–SR complex structure illustrates beautifully how SRP and SR function as each other's GAP and effector at the same time.^{5,6} To put it more generally, the biological effect (coat assembly/vesicle formation or biological complex formation) promoted by an active GTP-bound GTPase also functions as a GAP to inactivate the GTPase by promoting GTP hydrolysis, providing a spatial and temporal control of the GTPase function.

Summary

The GTP hydrolysis mechanism of Ras-like GTPases as well as a large number of other GTPases is of biological significance and of general interest and various models have been proposed over the last two decades.^{29,32,36,38} It is timely to review recent advances that provide insights into the GTP hydrolysis mechanism. While earlier models invoked associative transition state, recent studies have suggested the possibility that GTPase-catalyzed GTP hydrolysis reactions may proceed *via* a mostly dissociative transition state. This controversy is partly due to the different techniques used in the studies and the limitations of each technique when used alone. It is to be hoped that a combination of these techniques and application of additional techniques, e.g. the kinetic isotope effect that measures the charge difference between the transition state and the ground state, can resolve this difficult issue in the near future.

An emerging consensus is that GTPases catalyze

GTP hydrolysis reactions by neutralizing the negative charge development at β - or γ -phosphate of GTP and by correct positioning of the nucleophilic water molecule *via* a conserved Gln residue in the switch II region (or another mechanism in the cases of SRP and Rap1 systems). The low intrinsic GTPase activity perhaps reflects the relative weak catalytic groups such as the backbone amide groups in the P-loop as well as the incorrect geometry of the switch II Gln residue at ground state. The GAPs for Ras and Rho GTPases improve the reaction rate on both accounts.^{43,68,69} On one hand, GAPs provide a strong catalytic group in the form of an Arg finger that inserts into the nucleotide-binding pocket. On the other hand, GAPs stabilize the switch regions so that the Gln side-chain is in a geometry to position correctly the nucleophilic water molecule for attacking the γ -phosphate of GTP. The GAPs for Ran, ARF, and G_{α} subunits, however, enhance the reaction rate only by stabilizing the switch regions at the transition state,⁷⁹ suggesting that correct positioning of the nucleophilic water molecule relative to the γ -phosphate of GTP substrate is a more conserved mechanism for GTP hydrolysis. In this regard, the SRP–SR complex also provides a composite active site where the nucleophilic water molecule is positioned by a catalytic Asp residue for GTP hydrolysis,^{5,6} while Rap1GAP uses a catalytic Asn to position the nucleophilic water molecule.⁸⁰

The GTP hydrolysis reaction is critical for GTPases to work as molecular switches in the regulation of diverse cellular functions such as signal transduction, cytoskeleton organization, protein synthesis/translocation, and membrane trafficking. The studies on the basic GTP hydrolysis mechanism have contributed to our understanding of GTPase enzymology and of GTPase functions and mutations, e.g. constitutive active mutations are usually associated with defective GTP hydrolysis. GTPase mutants are useful tools in dissecting the cellular functions and an interesting challenge is to engineer a super GTPase with accelerated intrinsic GTP hydrolysis rate, which awaits a better understanding of the fundamental GTP hydrolysis mechanism.

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